LAB 2 TỔNG HỢP
Music Library Manager App

1. Project Overview

The **Music Library Manager Application** is a mobile application that helps users manage songs in a personal music library.

The application allows users to:

- login to the system (using hard-coded account)
- manage session state
- view songs
- add new songs
- edit songs
- delete songs
- search songs
- filter songs by genre or year
- mark songs as favorite
- view favorite songs

📌 **Important Constraints:**

- The application uses **ONLY 1 data table (Song)**
- User data is **NOT stored in database**
- Login uses **hard-coded username & password**
- No password encryption (no bcrypt required)

📌 All application data must be stored locally using:

- SQLite hoặc Hive

📌 The application must provide a:

- clean
- modern
- user-friendly interface

2. Learning Objectives

After completing this assignment, students should be able to:

- build multi-screen mobile apps with Flutter
- implement a simple login flow (hard-coded)
- manage user sessions using local storage
- perform CRUD operations with local database
- implement search and filter functionality
- manage favorite state within a single table
- design user-friendly mobile UI

3. Technology Requirements

Component	Requirement
Framework	Flutter
Language	Dart
Database	SQLite hoặc Hive
IDE	Android Studio / VS Code
UI Framework	Material Design

4. Project Naming

Students must use the following project name:

```
musicManagerBasic_StudentCode
```

Example:

```
musicManagerBasic_SE123456
```


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5. Data Model (Single Table Only)

Song Model

Field	Type	Description
id	int	Song ID (auto increment)
title	String	Song title
artist	String	Artist name
year	String	Release year
genre	String	Music genre
isFavorite	int (0/1)	Favorite status

📌 Notes:

- `isFavorite = 1` → favorite
- `isFavorite = 0` → normal

6. Application Screens

The application must include the following screens:

1️⃣ Login Screen
2️⃣ Home Screen (Song List)
3️⃣ Add Song Screen
4️⃣ Song Detail Screen
5️⃣ Edit Song Screen
6️⃣ Favorite Songs Screen

7. Functional Requirements

Task 1 – Login (Hard-coded)

Create a **Login Screen**

Fields:

- Username
- Password

👉 Use the following account:

```
Username: admin
Password: 123456
```

Requirements:

- Do NOT use database for login
- Do NOT encrypt password
- Validate input fields
- Compare directly in code

Example logic:

```
if (username == "admin" && password == "123456") {
   // login success
}
```

If login succeeds:

- Save login session
- Navigate to Home Screen

If login fails:

❌ Display message:

```
Invalid username or password
```

Task 2 – Session Management

The application must remember login state.

When the app starts:

- Check session storage

If session exists:

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→ Open Home Screen
If session does not exist:
→ Open Login Screen
📌 Suggested tools:
• SharedPreferences

Task 3 – Logout

Add Logout button on Home Screen.
When clicked:

- Clear session data
- Navigate to Login Screen
- Show message:
  `Logged out successfully`

Task 4 – Home Screen (Song List)

Display a list of songs.
Display:

- Title
- Artist
- Year

Requirements:

- Load data from database
- Use ListView.builder

Navigation:

- Add Song
- Song Detail
- Favorite Songs
- Logout

Task 5 – Add Song

Create Add Song Screen.
Fields:

- Title
- Artist
- Year
- Genre

Requirements:

- ID must auto-increment
- `isFavorite = 0` by default
- Save to database

After saving:

- Return to Home Screen
- Update song list

Task 6 – Song Detail

When user selects a song:
Display:

- Title
- Artist
- Year
- Genre

Actions:

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- Edit Song
- Delete Song
- Toggle Favorite (❤️)

Task 7 – Update Song

Users can edit song information.
Requirements:

- Modify any field
- Save to database
- Update immediately on Home Screen

Task 8 – Delete Song

Add Delete button on Song Detail Screen.
Before deletion:
⚠️ Show confirmation dialog
Example:

```
Are you sure you want to delete this song?
```

After deletion:

- Remove from database
- Update list
- Show SnackBar notification

Task 9 – Search Songs

Add a Search Bar on Home Screen.
Requirements:

- Search by Title or Artist
- Filter results in real-time

Example:
Search: *Love*
→ Love Story
→ Love Yourself

Task 10 – Filter Songs

Allow filtering by:
🎵 Genre (e.g., Pop, Rock, EDM, Ballad)
📅 Year:

- Current year
- Last 5 years
- All

Suggested UI:

- DropdownButton
- Filter Chips

Task 11 – Favorite Songs

Users can mark/unmark songs as favorite.
Implementation:

- Update `isFavorite` field

Example:

```
isFavorite = 1 → favorite
isFavorite = 0 → normal
```

Requirements:

- Toggle favorite status
- Update UI immediately

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Task 12 – Favorite Songs Screen

Create Favorite Songs Screen.
Display only songs where:

```
isFavorite = 1
```

Users can:

- Remove from favorites

8. UI Requirements

The application should:

- Use Material Design
- Be clean and easy to use
- Have consistent layout
- Provide user feedback (SnackBar, Dialog)

9. Submission Requirements

Students must submit:

- Source code (Flutter project)
- Screenshots of all screens
- Short description of features implemented

10. Bonus (Optional)

Students may implement:

- Sort songs (by name, year)
- Dark mode
- Add album image (local)
- Simple animations

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NOTES

Music Manager (Quản lý bài hát)

📌 Có các chức năng chính:

1. Login (hard-code)
2. Lưu trạng thái đăng nhập
3. CRUD bài hát (Thêm / Sửa / Xóa)
4. Tìm kiếm + lọc
5. Đánh dấu yêu thích (favorite)

📌 Lưu ý quan trọng:

- ❌ Không có bảng User
- ❌ Không có nhiều bảng
- ✅ Chỉ có 1 bảng: **Song**
- ✅ Login viết if cứng (admin / 123456)

🧭 2. Lộ trình làm bài (RẤT QUAN TRỌNG)

👉 Làm theo thứ tự này để tránh rối:

Bước 1
Tạo project Flutter
Bước 2
Làm Login (hard-code)
Bước 3
Làm Session (SharedPreferences)
Bước 4
Tạo Database + Model Song
Bước 5
Làm Home Screen (hiển thị list)
Bước 6
Làm chức năng CRUD
Bước 7
Làm Favorite
Bước 8
Làm Search + Filter

🛠️ 3. Hướng dẫn từng bước

🔷 Bước 1: Tạo project

```
npx create-expo-app (KHÔNG dùng ❌)
flutter create musicManagerBasic_xxx (ĐÚNG ✅)
```

Mở bằng VS Code → chạy:

```
flutter run
```

🔷 Bước 2: Tạo Login Screen

📌 Tạo file:

```
login_screen.dart
```

📌 UI gồm:

- TextField Username
- TextField Password
- Button Login

📌 Logic:

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```
if(username == "admin" && password == "123456"){
   // thành công
}else{
   // báo lỗi
}
```

👉 Nếu đúng:
→ chuyển sang Home Screen
👉 Nếu sai:
→ show: *Invalid username or password*

🔷 Bước 3: Session (Ghi nhớ đăng nhập)

📌 Dùng:
`SharedPreferences`

👉 Khi login thành công:
```
prefs.setBool("isLogin", true);
```

👉 Khi mở app:
```
if(isLogin == true){
   vào Home
}else{
   vào Login
}
```

👉 Logout:
```
prefs.remove("isLogin");
```

🔷 Bước 4: Tạo Model Song

📌 File:
`song.dart`

📌 Thuộc tính:
```
class Song {
  int? id;
  String title;
  String artist;
  String year;
  String genre;
  int isFavorite;
}
```

🔷 Bước 5: Tạo Database

👉 Chọn 1 trong 2:
• SQLite (khuyên dùng)
• Hive

📌 Nếu dùng SQLite:
• tạo bảng:
```
Song(
 id INTEGER PRIMARY KEY AUTOINCREMENT,
 title TEXT,
 artist TEXT,
 year TEXT,
 genre TEXT,
 isFavorite INTEGER
)
```


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🔷 Bước 6: Home Screen (Danh sách bài hát)

📌 Dùng:

```
ListView.builder()
```

Hiển thị:

• title
• artist
• year

📌 Khi click:

→ sang Detail Screen

🔷 Bước 7: Add Song

📌 Tạo form:

• title
• artist
• year
• genre

📌 Khi save:

```
isFavorite = 0
```

👉 Sau khi lưu:

→ quay lại Home
→ reload list

🔷 Bước 8: Detail + Edit + Delete

Detail Screen:

Hiển thị full info

Edit:

• cho sửa lại field
• update database

Delete:

📌 Hiện confirm:

```
Are you sure?
```

👉 Nếu OK:

• xóa DB
• quay lại Home
• show SnackBar

🔷 Bước 9: Favorite

👉 Không cần bảng mới ❗

📌 Toggle:

```
isFavorite = 1 // yêu thích
isFavorite = 0 // bình thường
```

👉 UI:

❤️ icon

🔷 Bước 10: Favorite Screen

📌 Query:

```
SELECT * FROM Song WHERE isFavorite = 1
```

Hiển thị danh sách favorite

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🔷 Bước 11: Search
📌 TextField trên Home
👉 Logic:
`song.title.contains(keyword)`
👉 Làm real-time (onChanged)

🔷 Bước 12: Filter
📌 Lọc theo:
• Genre
• Year
👉 Gợi ý:
• Dropdown
• Chips

⚠️ 4. Lỗi sinh viên hay gặp
❌ Quên reload list sau khi add/update/delete
❌ Không set isFavorite mặc định = 0
❌ Sai logic login (thiếu &&)
❌ Không lưu session → mở app lại bị logout
❌ Không await async DB

💡 5. Mẹo để làm nhanh
👉 Làm từng phần nhỏ:
1. Login chạy OK ✅
2. Session chạy OK ✅
3. CRUD chạy OK ✅
4. Sau đó mới làm UI đẹp

👉 Không làm tất cả cùng lúc ❗

🧠 6. Tư duy khi làm bài
Bạn chỉ cần nhớ:
👉 App này thực chất chỉ là:
• 1 list dữ liệu (Song)
• + thêm/sửa/xóa
• + lọc

🎯 7. Checklist hoàn thành
✓ Login đúng
✓ Có session
✓ Thêm bài hát
✓ Sửa bài hát
✓ Xóa bài hát
✓ Favorite hoạt động
✓ Search chạy
✓ Filter chạy

Phucpt10@fpt.edu.vn